Transcription & Nuclear Hormone Receptors

Charles P. Scott, PhD

(charles.scott@jefferson.edu)

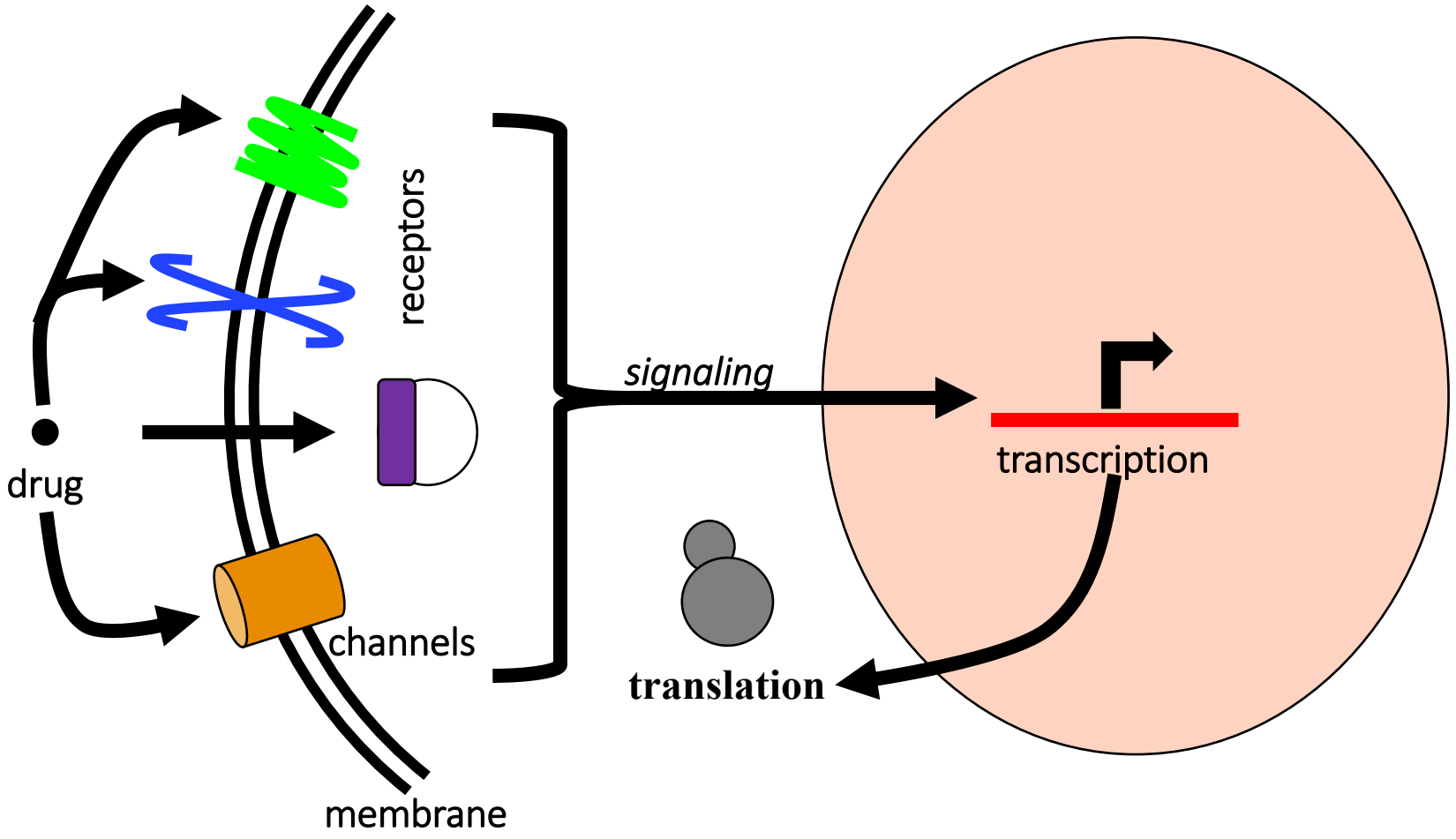
JeffMD Block 5

October 13th, 2026

Modified from a presentation given by Dr. Diane Merry on 8/9/22

Dr. Scott has no conflicts of interest to declare

Membrane permeable signaling molecules activate intracellular receptors



Nuclear Hormone Receptors

Transcriptional regulators whose activity is activated by small ligands

- Transcriptional regulators are proteins that bind to DNA at specific sites and cause a change in the transcription of a gene or set of genes – new RNA made.

Ligands include steroids, retinoic acid, thyroid hormone and vitamin D3

Regulate diverse developmental and physiological processes

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7. Outline the mechanism of action of peroxisome proliferator-activated receptors (PPARs) and discuss the utility of these receptors in the management of dyslipidemia and insulin resistance.

ScholarRx 39305.8

A 54-year-old woman who has entered postmenopause comes to the clinic for vaginal dryness, hot flashes, and dyspareunia. She begins estrogen-progestin hormone replacement therapy. After one month of treatment, which set of values in the chart would best reflect the patient's thyroid status?

Choice	Thyroxine (T_4)-binding globulin	Free T_4	Free triiodothyronine (T_3)	Total T_3	Thyroid-stimulating hormone (TSH)
A	Normal	↑	↑	↑	↓
B	↑	Normal	Normal	↑	Normal
C	Normal	↓	↓	↓	↑
D	Normal	Normal	↓	↓	Normal
E	↓	Normal	Normal	↓	Normal

P value: 0.53

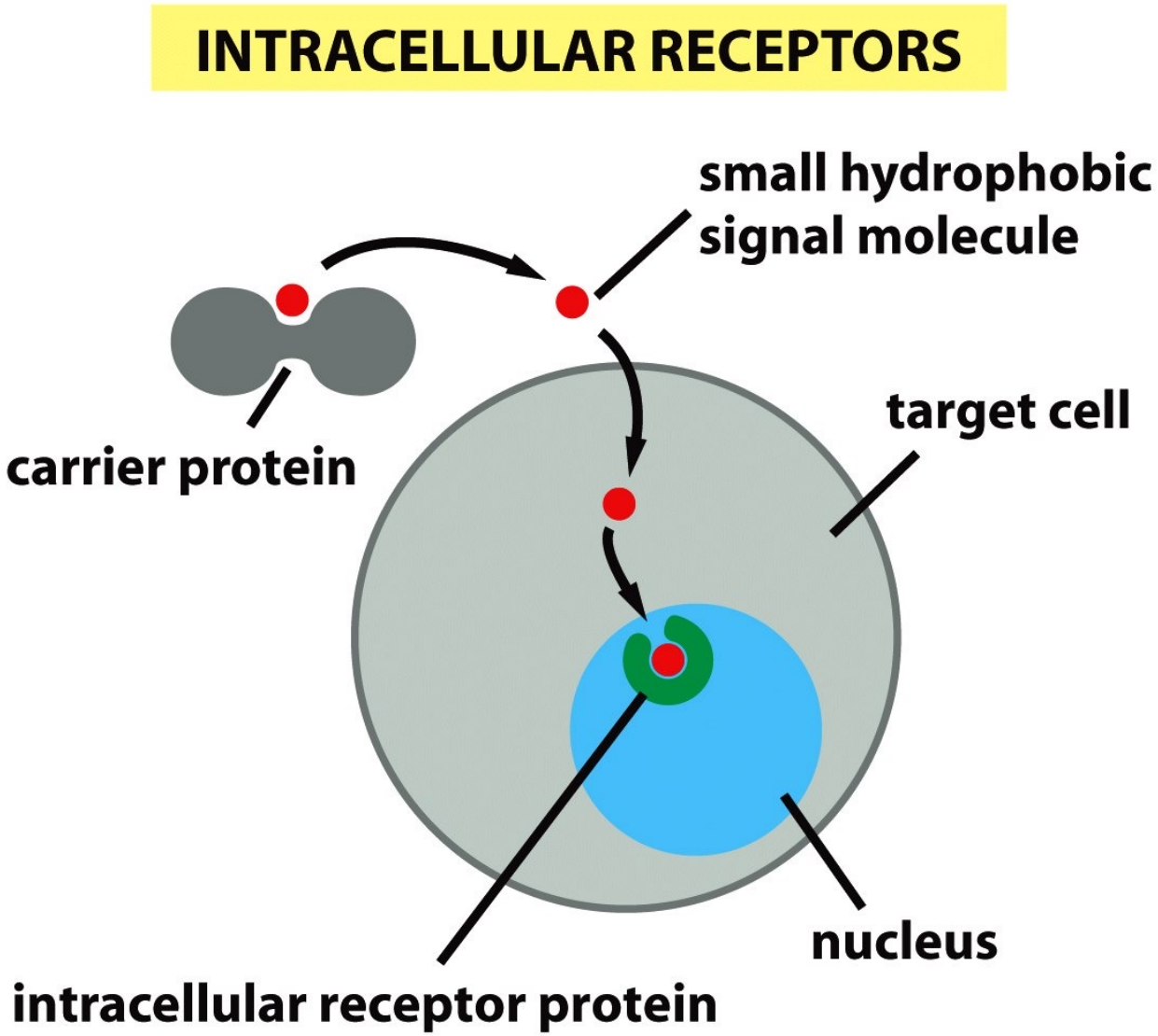


Figure 15-3b *Molecular Biology of the Cell* (© Garland Science 2008)

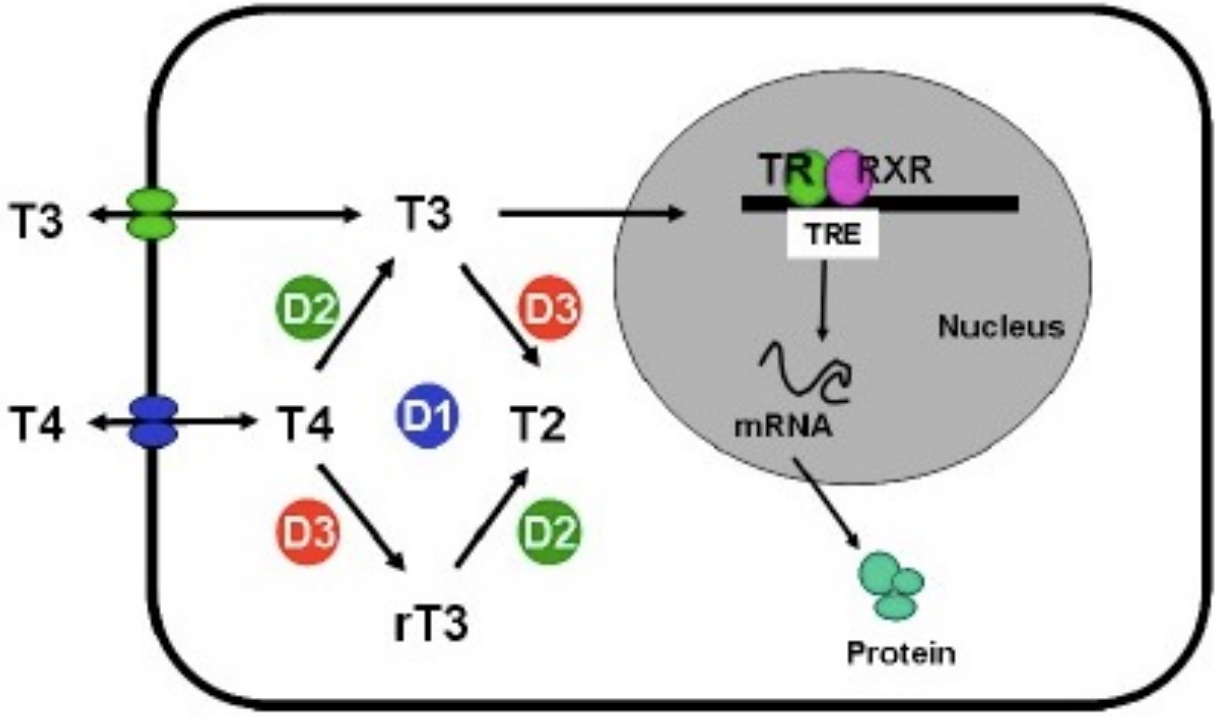
Steroid hormones enter cells by passive diffusion

Freely diffuse across the cell membrane.

No evidence for uptake in protein-bound form.

Within the cell, steroid hormones may be converted to other forms by metabolizing enzymes (e.g., aromatase, 5-α-reductase)

T4 is transported in the blood by **thyroxine binding globulin** and enter cells using specific transporter proteins; at least 10 transporters identified to date, including **organic anion transporting polypeptides (OATPs)** and amino acid transporters, including **monocarboxylate transporters** MCT8 and MCT10.



https://www.ncbi.nlm.nih.gov/books/NBK285565/

Visser TJ. Cellular Uptake of Thyroid Hormones. [Updated 2016 Jun 15]. In: De Groot LJ, Chrousos G, Dungan K, et al., editors. Endotext [Internet]. South Dartmouth (MA): MDText.com, Inc.; 2000-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK285565/

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ScholarRx 43443.1

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B. Lysine

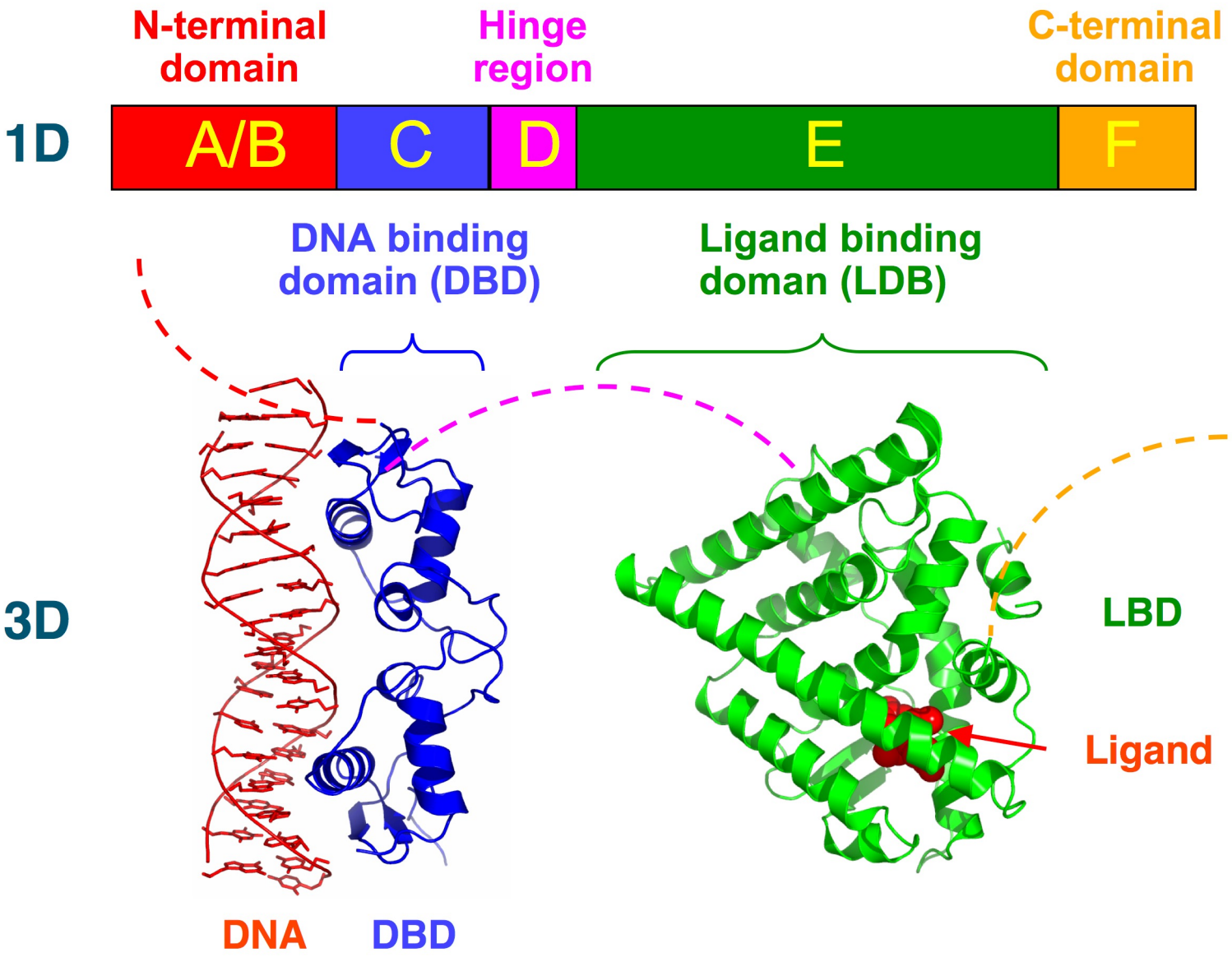
C. Proline

D. Serine

E. Threonine

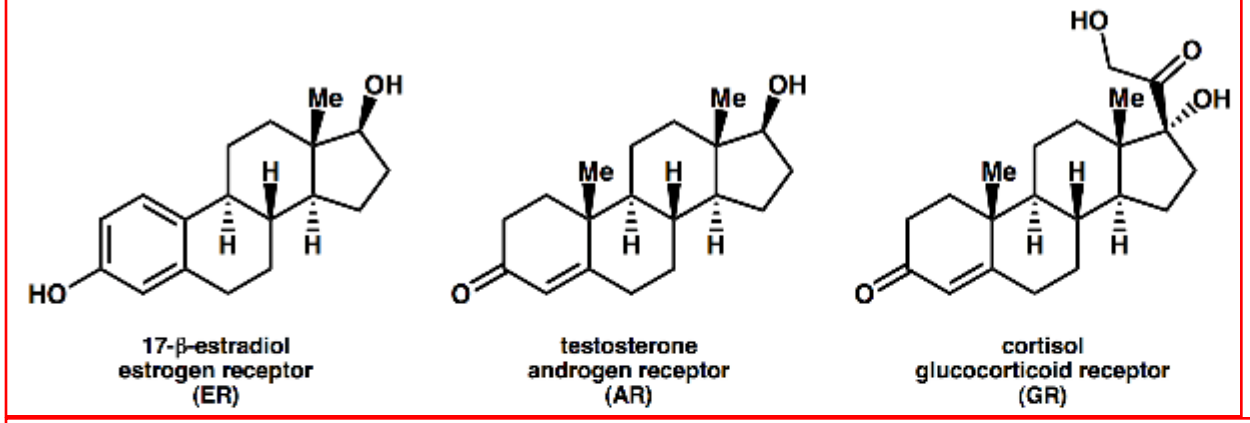
P value 0.45

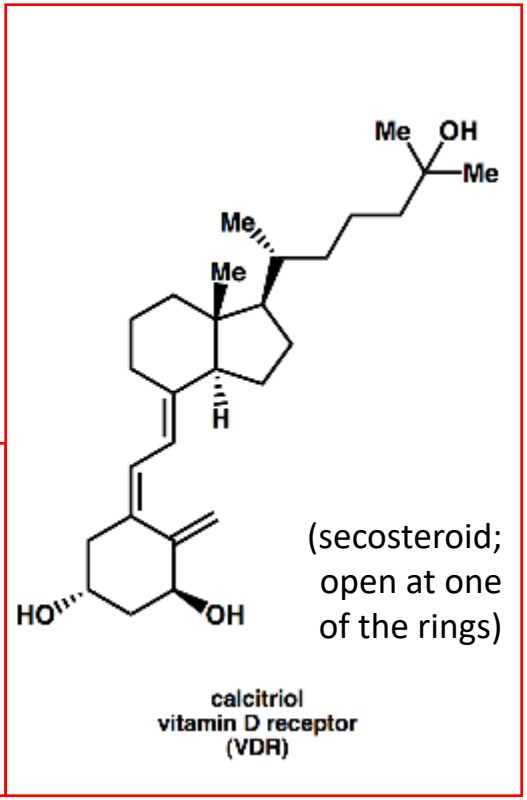
Structural Organization of Nuclear Receptors

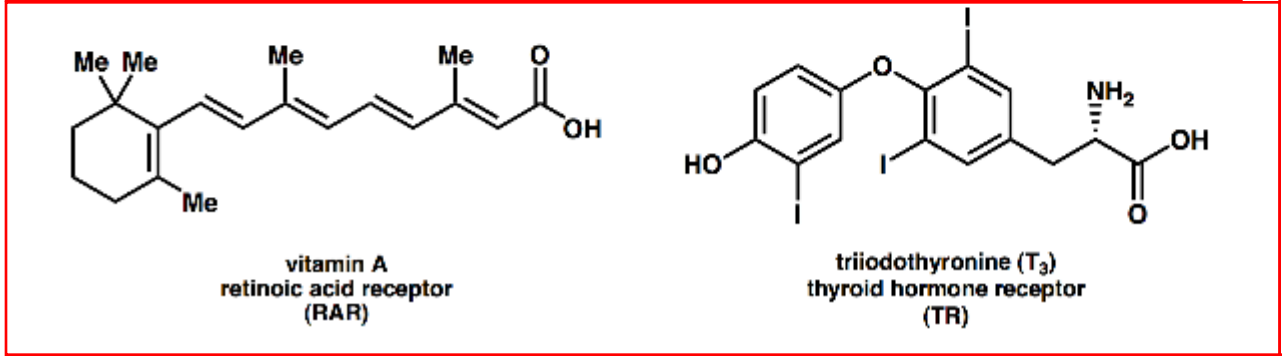


Type I & II NHRs recognize different classes of ligands

Steroidal hormone ligands (Type I NHRs)



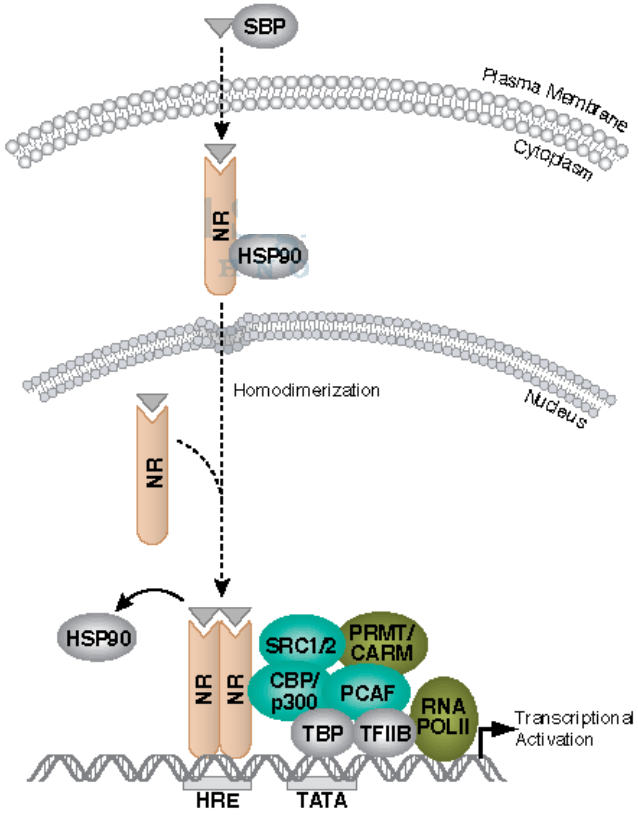




Non-steroidal hormone ligands (Type II NHRs)

Type I & II NHRs regulate transcription differently

Type I Steroid Receptors (**Homodimers**)



Type II Non-steroid Receptors (RXR **Heterodimers**)

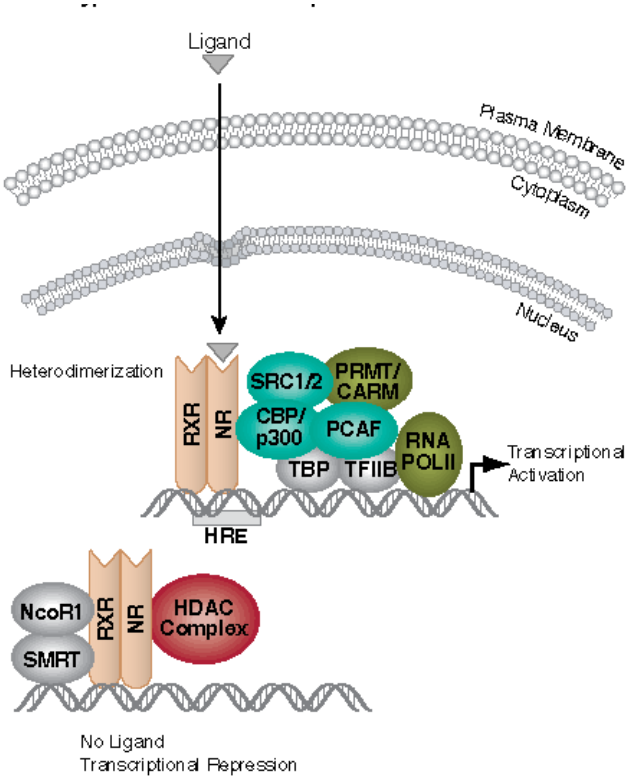
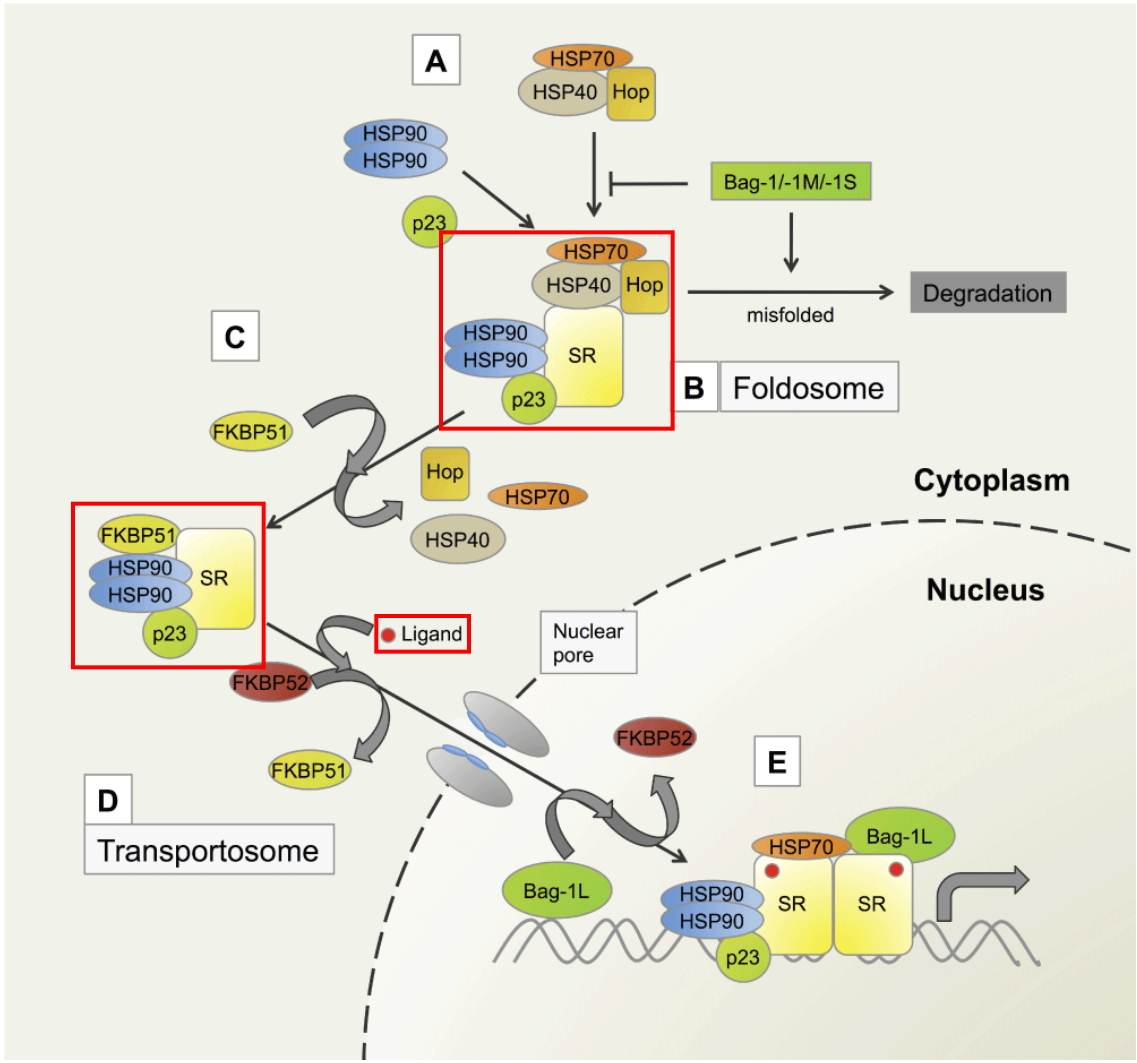


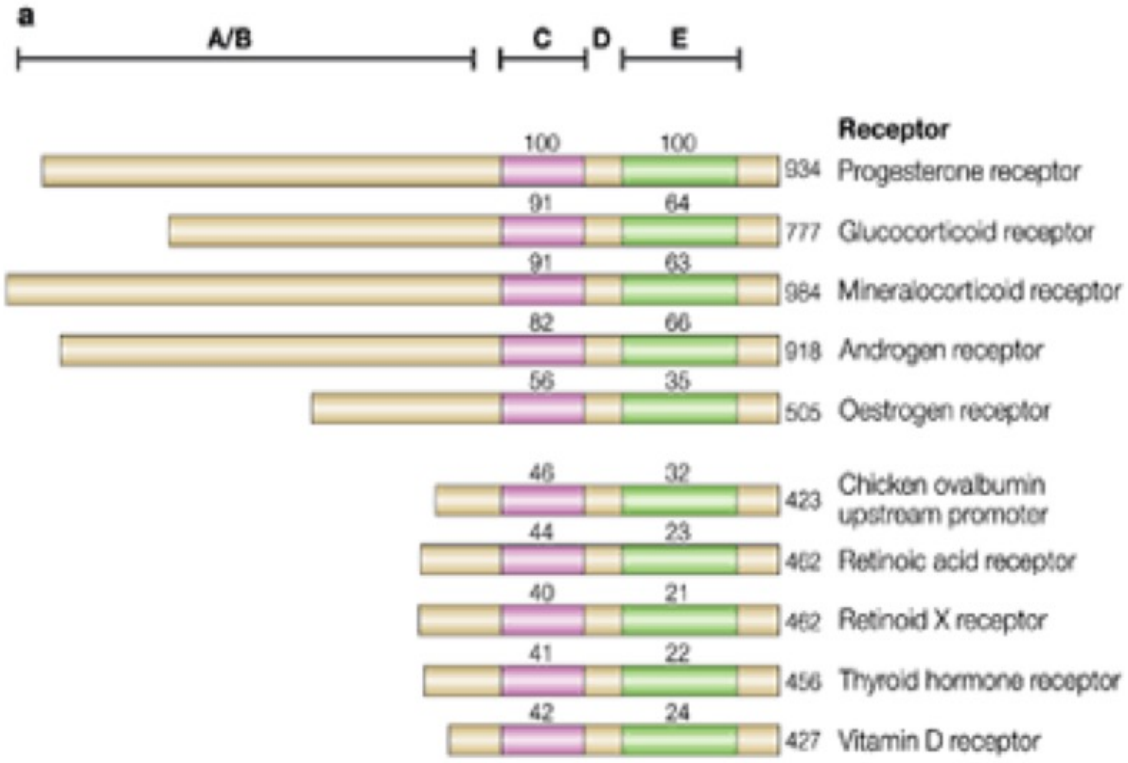
Illustration reproduced courtesy of Cell Signaling Technology, Inc. (www.cellsignal.com)

The role of molecular chaperones in the folding and transport of SHRs



Cato et al. (2014) Nucl Recept Signal 12, e005

General structure of the NHR superfamily



Type I

Type II

ScholarRx 43443.1

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ScholarRx 43054

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A. Enzyme-linked receptor

B. G protein-coupled receptor

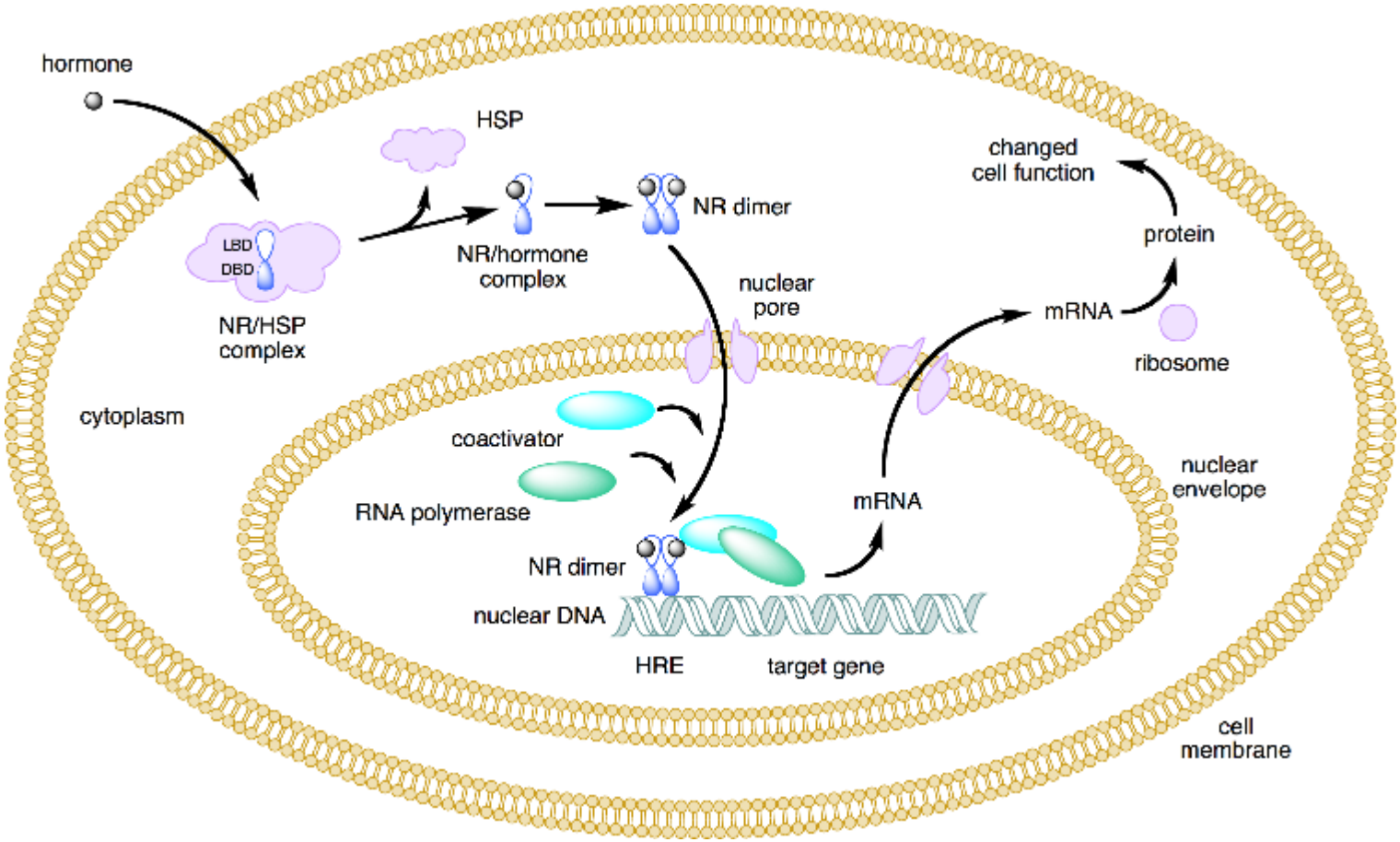
C. Intracellular receptor

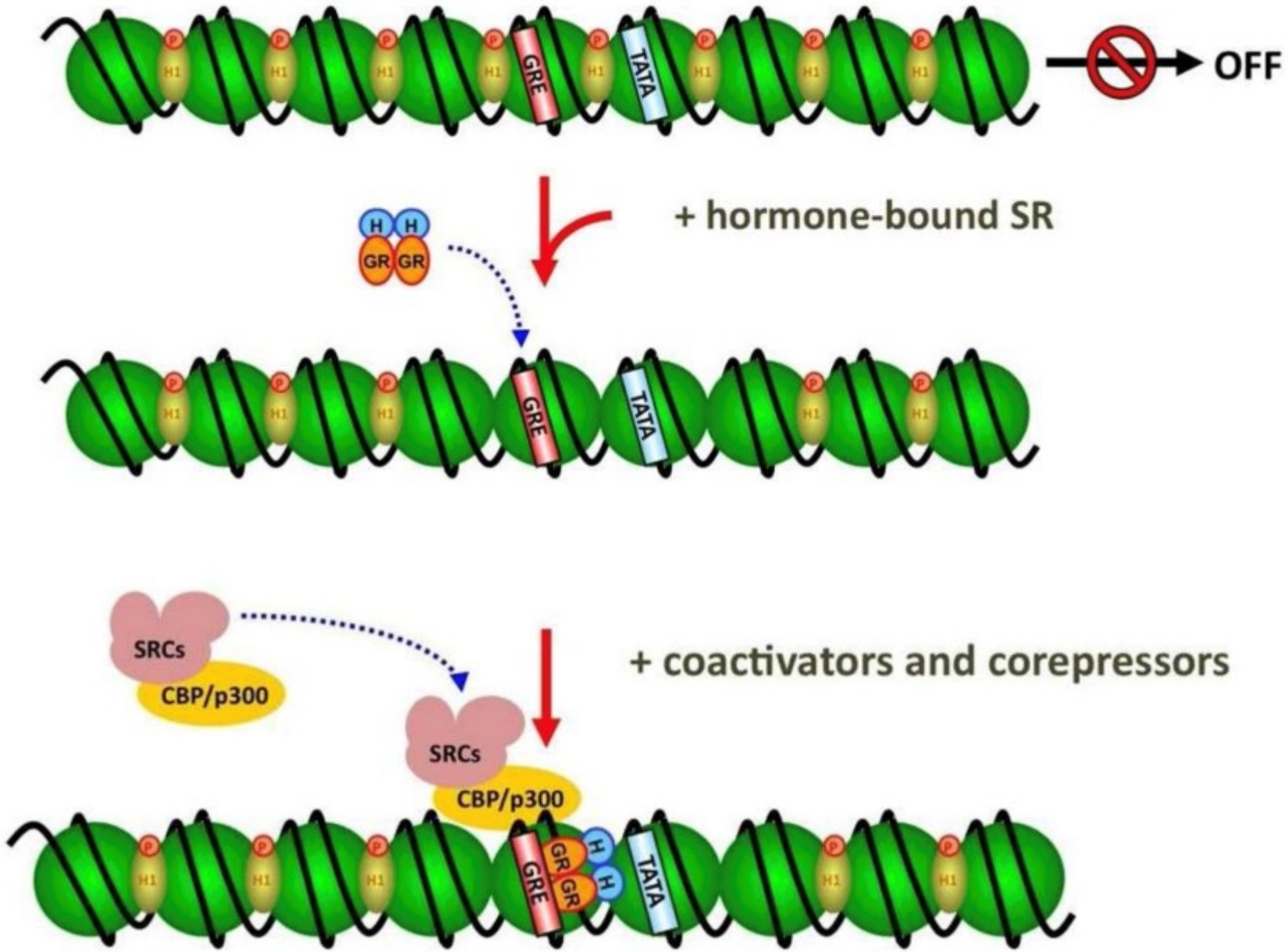
D. Ionotropic receptor

E. Mechanoreceptor

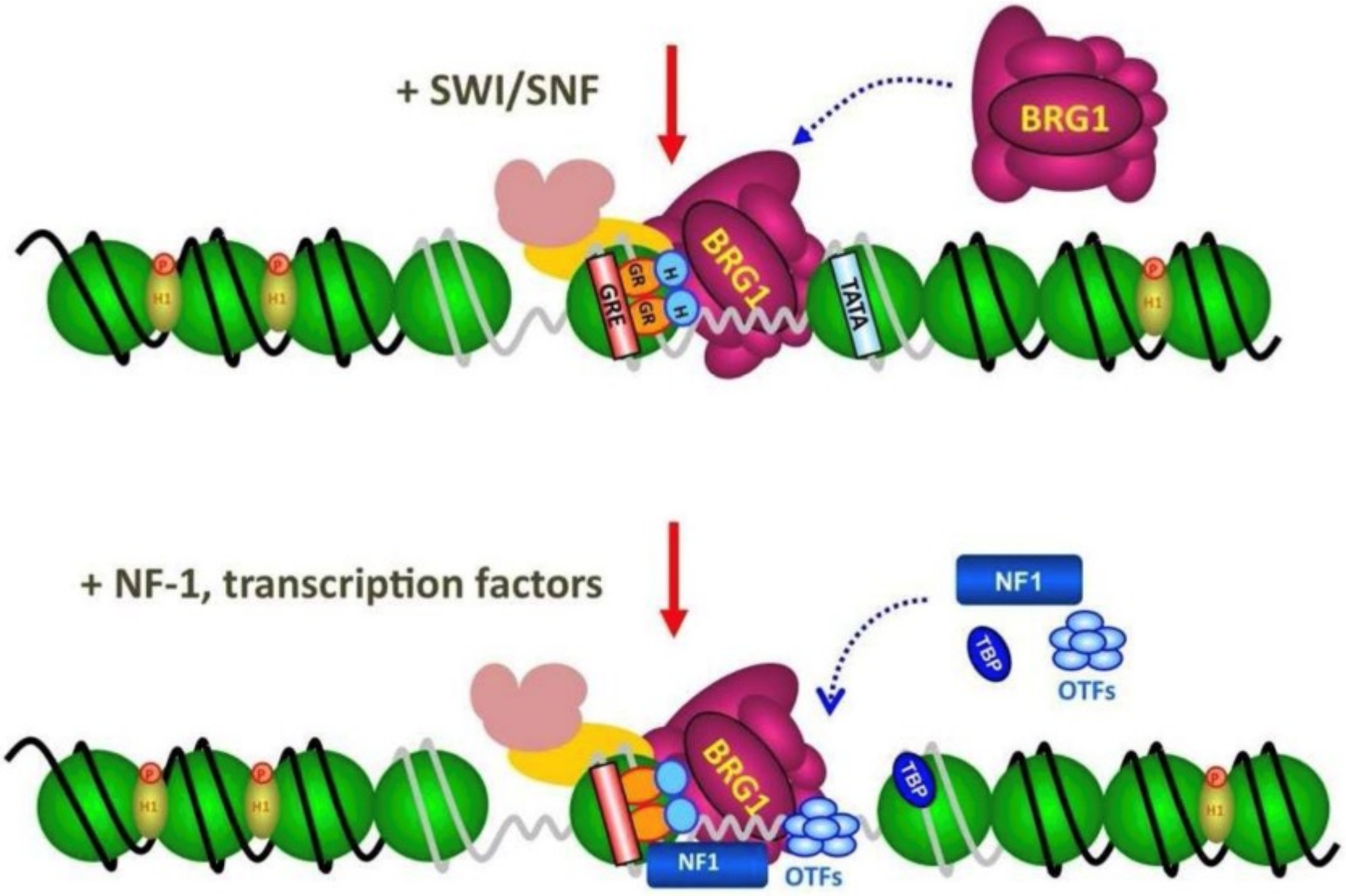
P value: 0.69

Mechanism of action of Type I nuclear hormone receptors: glucocorticoid receptor, mineralocorticoid receptor, androgen receptor, progesterone receptor, estrogen receptor

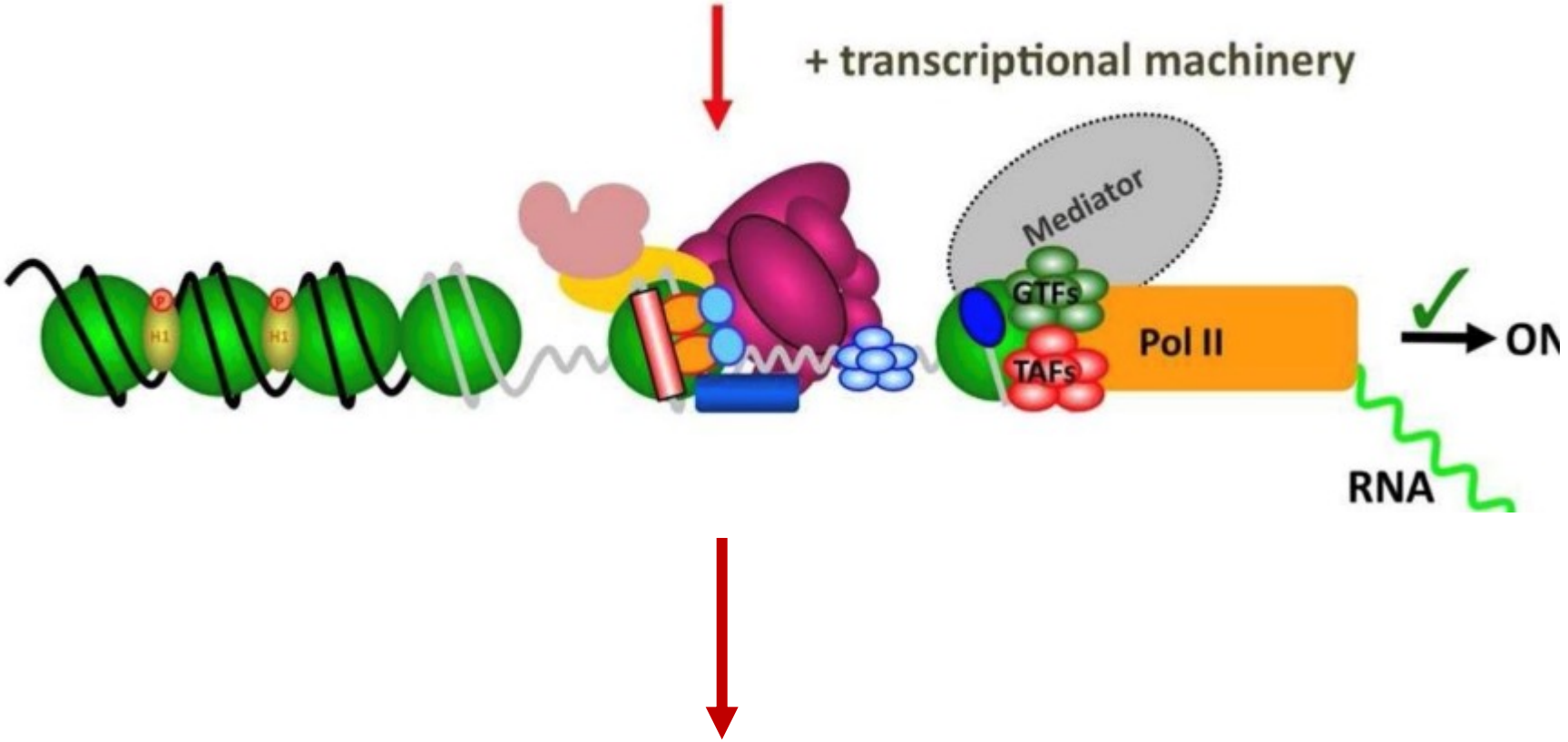




King et al., Biochimica et Biophysica Acta (BBA) - Gene Regulatory Mechanisms, Volume 1819, Issue 7, 2012, 716–726



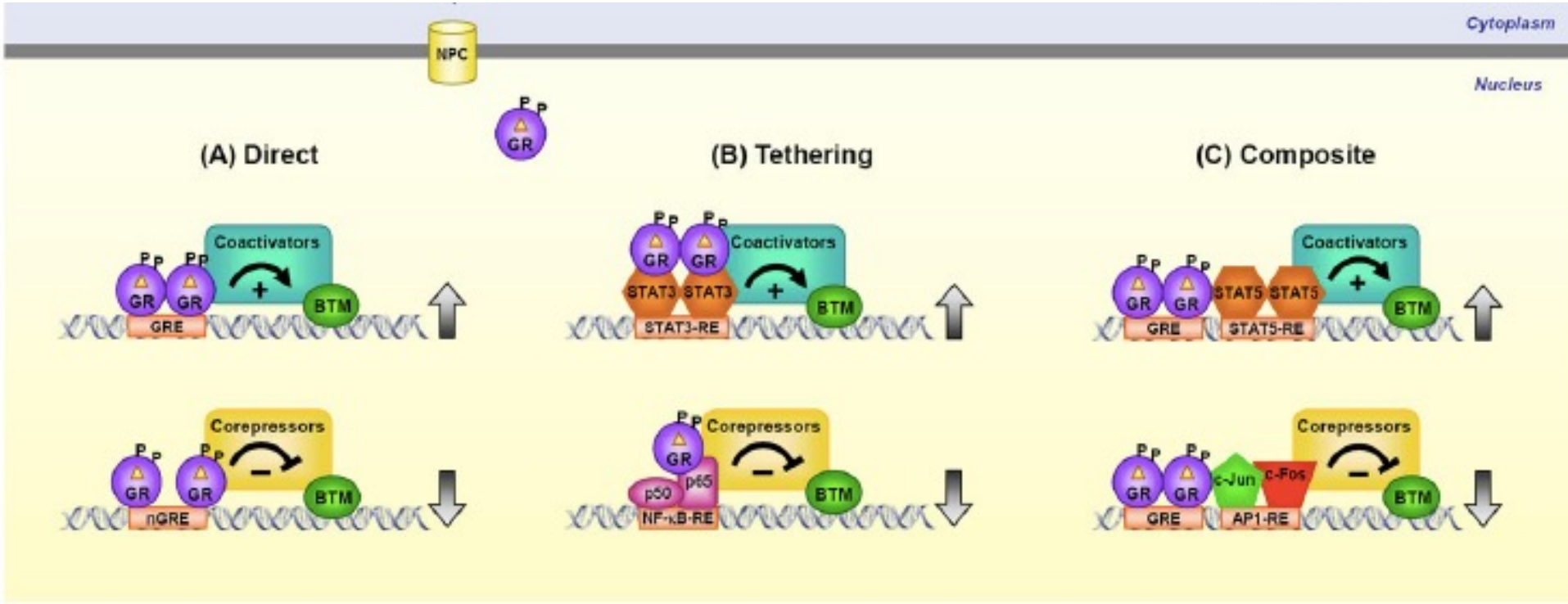
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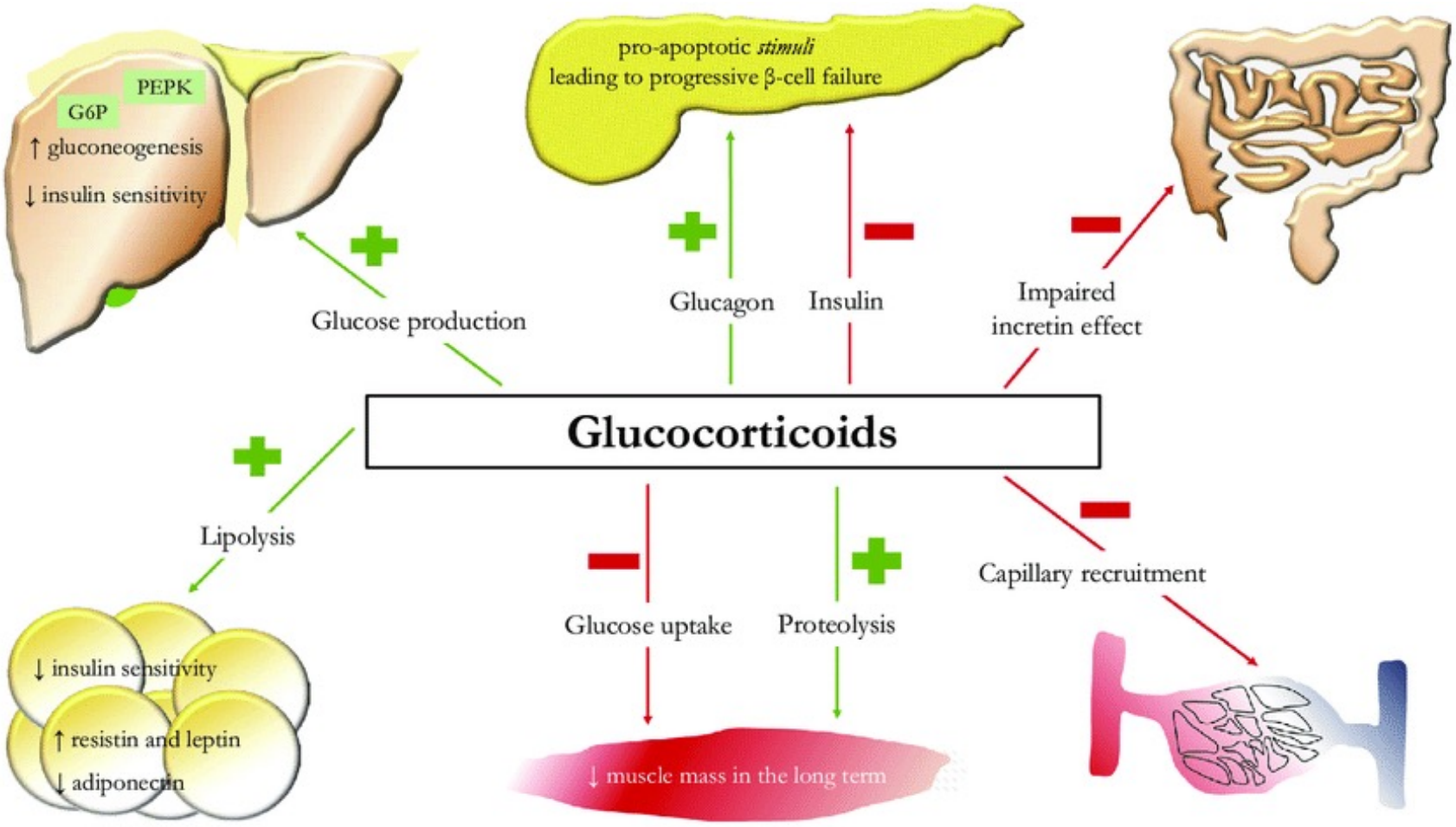
Occurs at thousands of DNA sites leading to hundreds of new proteins produced

King et al., Biochimica et Biophysica Acta (BBA) - Gene Regulatory Mechanisms, Volume 1819, Issue 7, 2012, 716–726

Regulation of steroid hormone (Type I) NHR

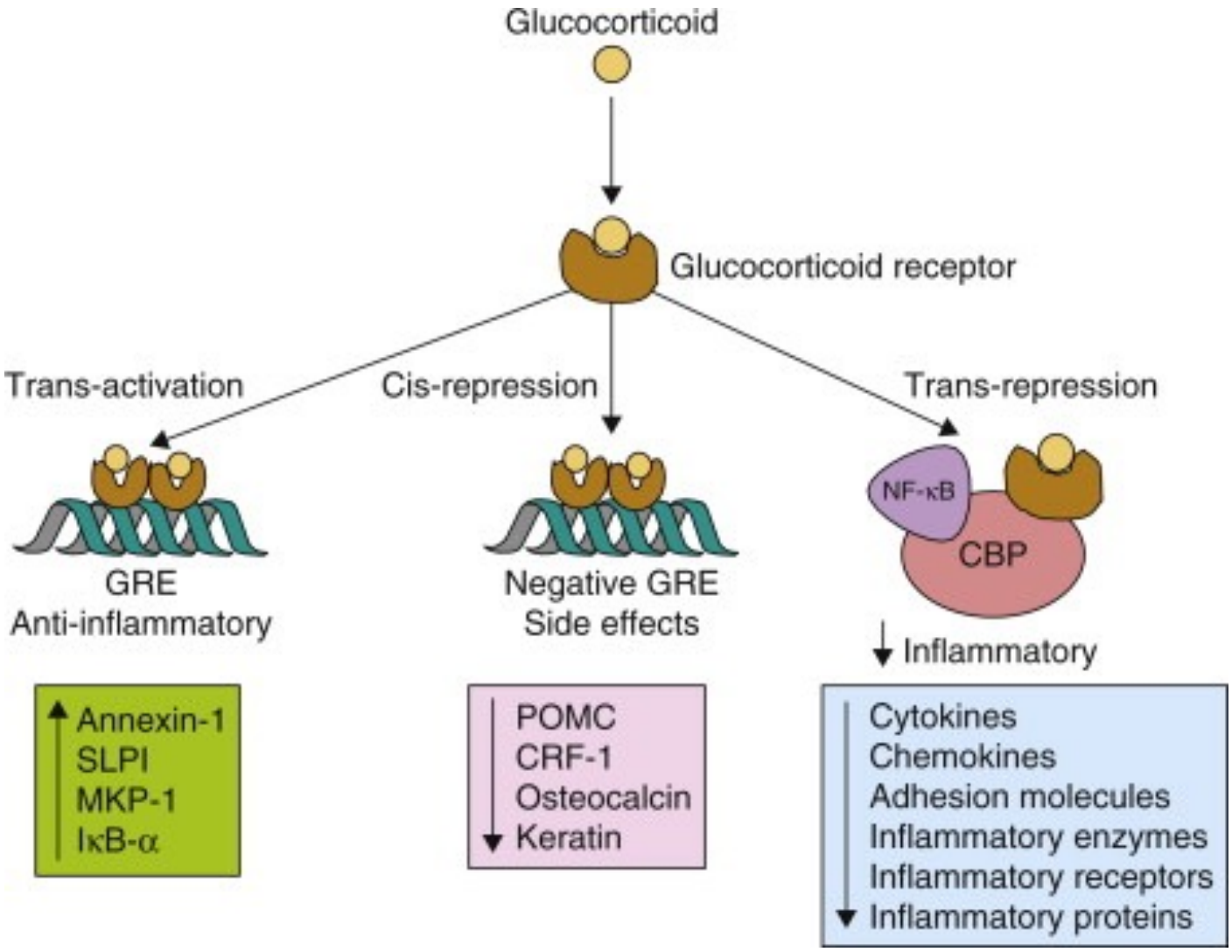


Glucocorticoids are named for their metabolic effects

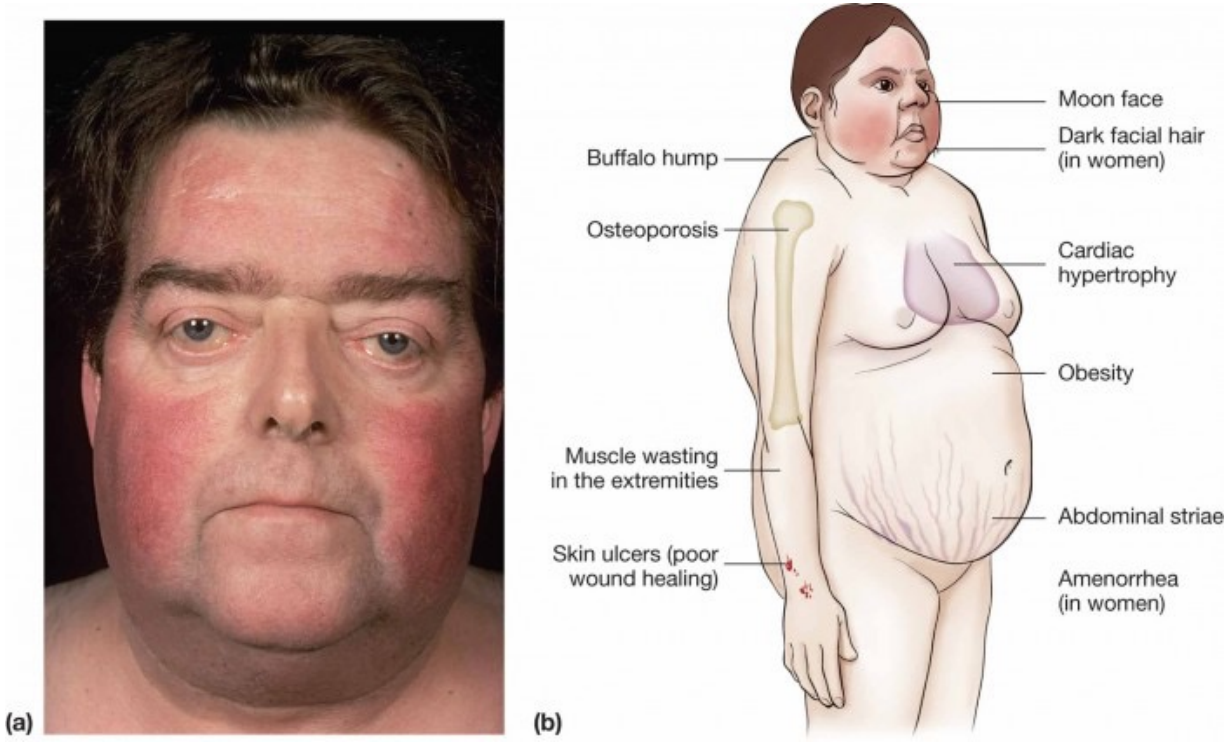


Diabetes Research and Clinical Practice (2018) DOI: 10.1016/j.diabres.2018.03.006

…but prescribed for anti-inflammatory effects



Cushing's syndrome



Symptoms reversible

Mostly iatrogenic

Endogenous treated surgically

Mifepristone (RU486) and ketoconazole used when inoperable

ScholarRx 43054

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A 4-month randomized, double blind controlled trial was conducted among 84 healthy women aged 17-50 years old. At baseline and 4 months after intervention, serum concentrations of TSH, total thyroxine (T4), total triiodothyronine (T3), and transthyretin (TTR) were measured. Which of the following reduced the risk of subclinical hypothyroidism in premenopausal women?

A. Vitamin A

B. Vitamin C

C. Vitamin D

D. Vitamin E

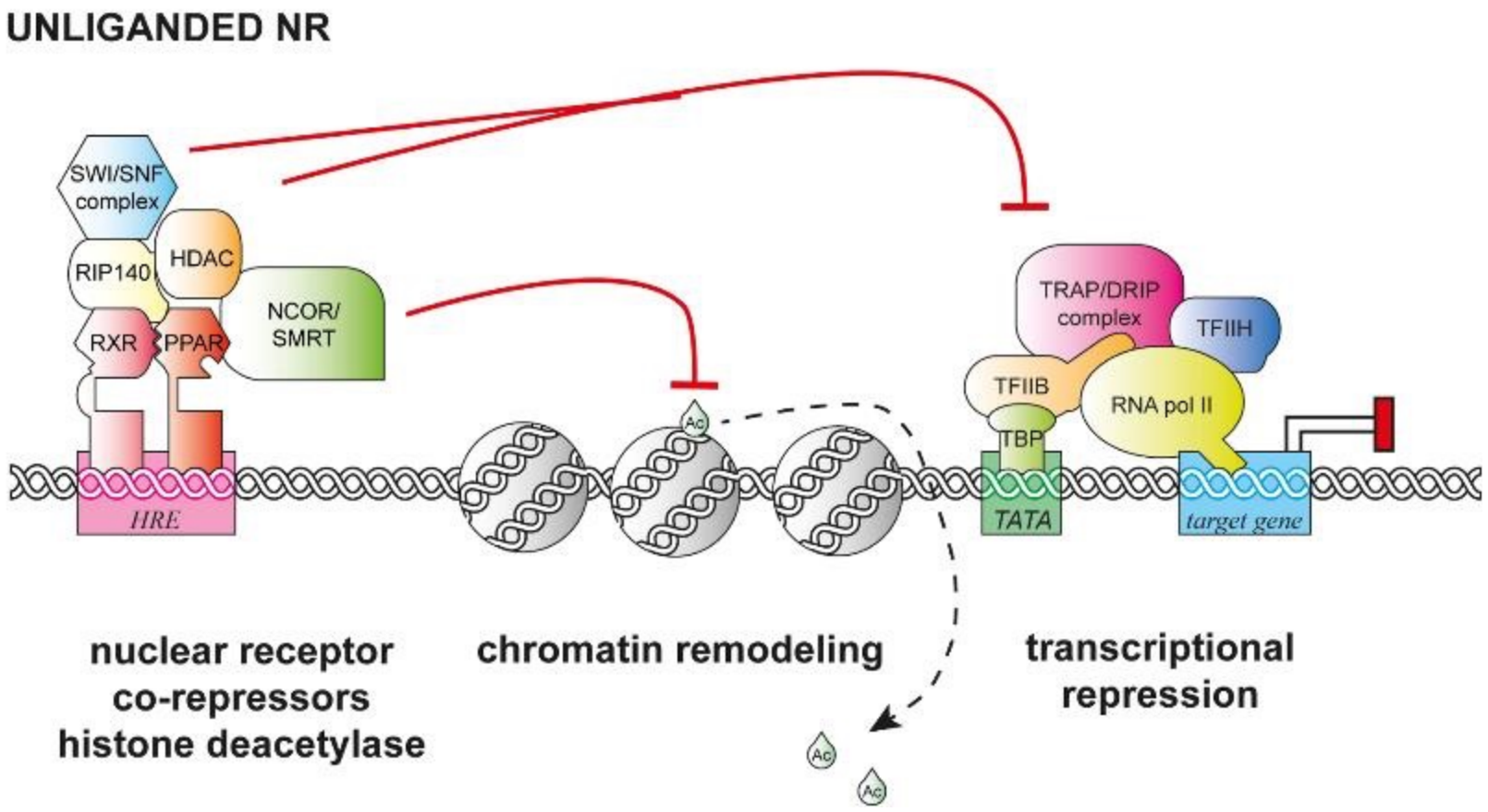
E. Vitamin K

DOI: 10.1080/07315724.2012.10720431

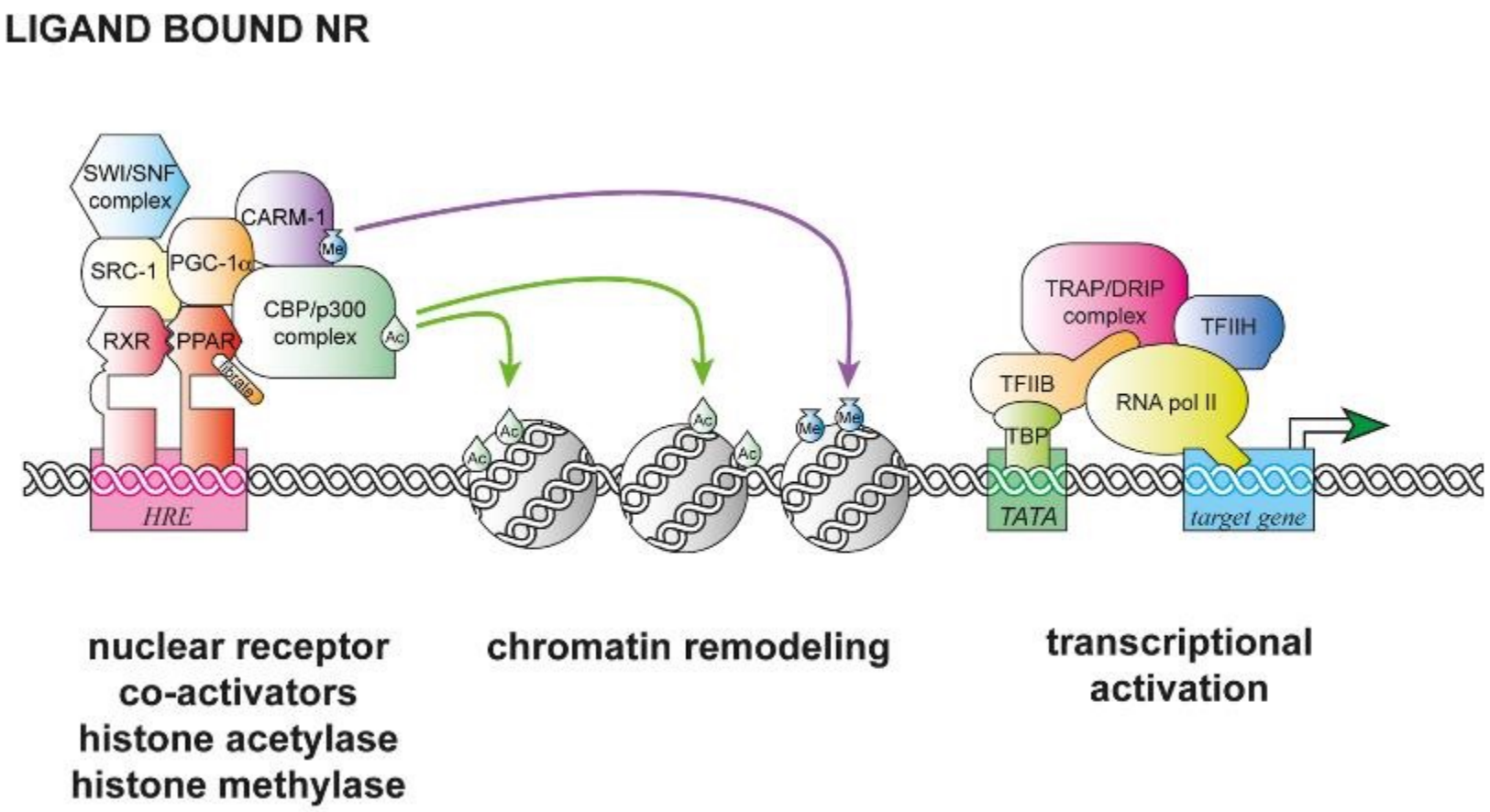
Type II NHR Mechanism of Action

- Type II nuclear receptors form heterodimers with retinoid-X receptor (RXR).
- Bind direct repeats in DNA
- Nuclear and DNA-bound in absence of hormone.
- Associated with **co-repressors** in absence of hormone.
- Hormone binding causes conformational change, release of co-repressors and recruitment of co-activators.

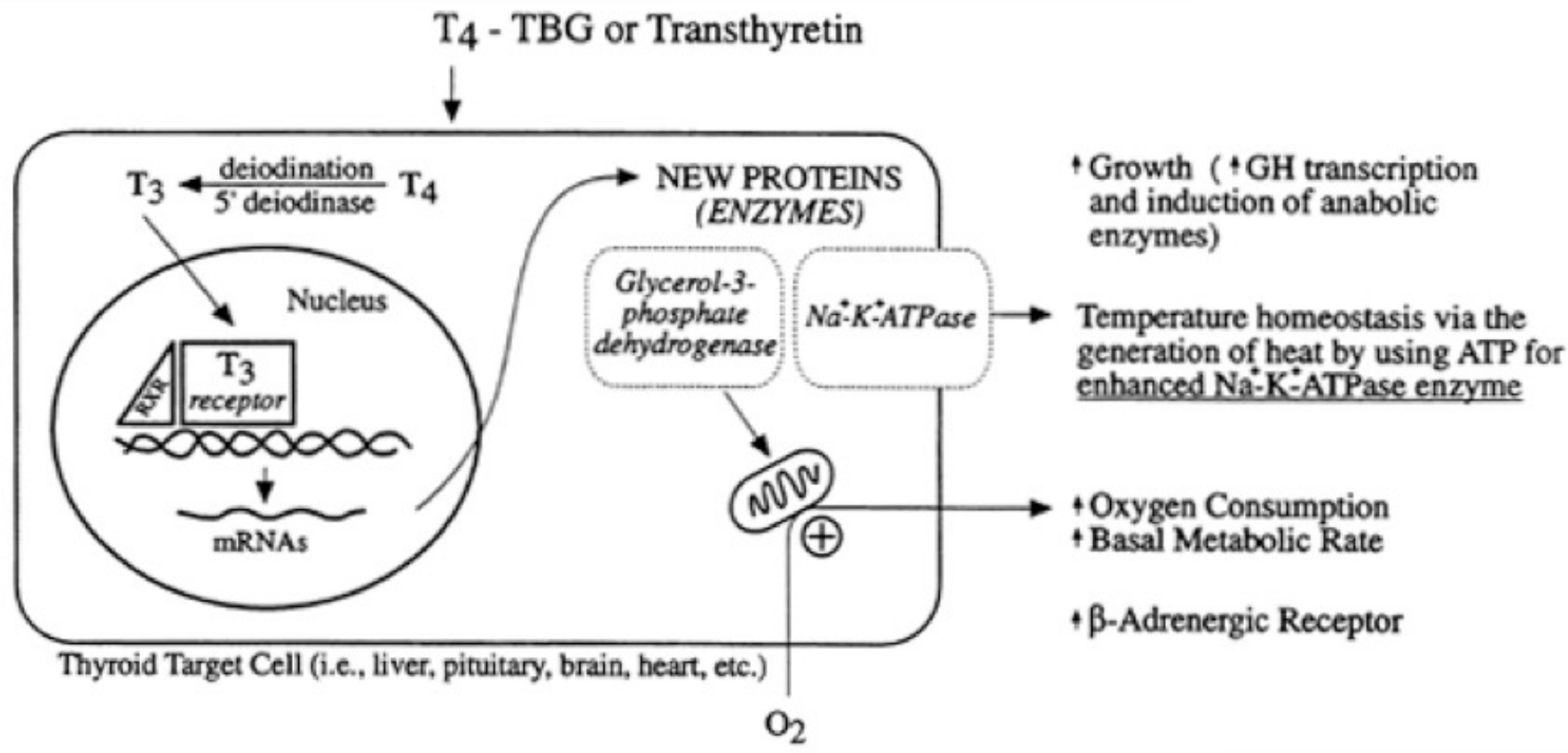
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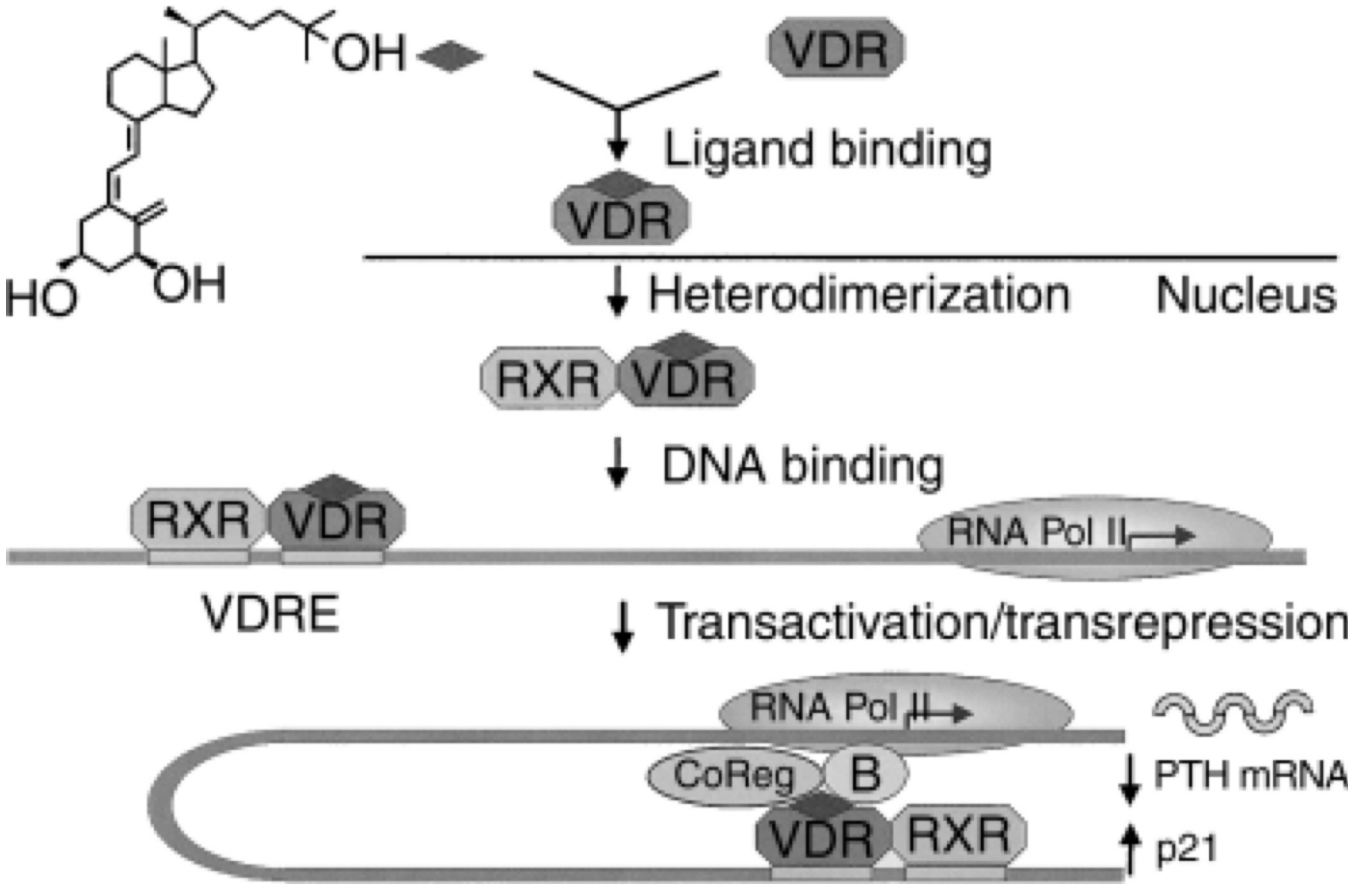
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Examples of type II NHRs: Thyroid receptor (TR)



Examples of class II NHRs: Vitamin D receptor (VDR)



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ScholarRx 2366.18

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B. Hyperkalemia and metabolic acidosis

C. Hyperstimulation of the ovaries

D. Inhibition of osteoclasts

E. Myelosuppression

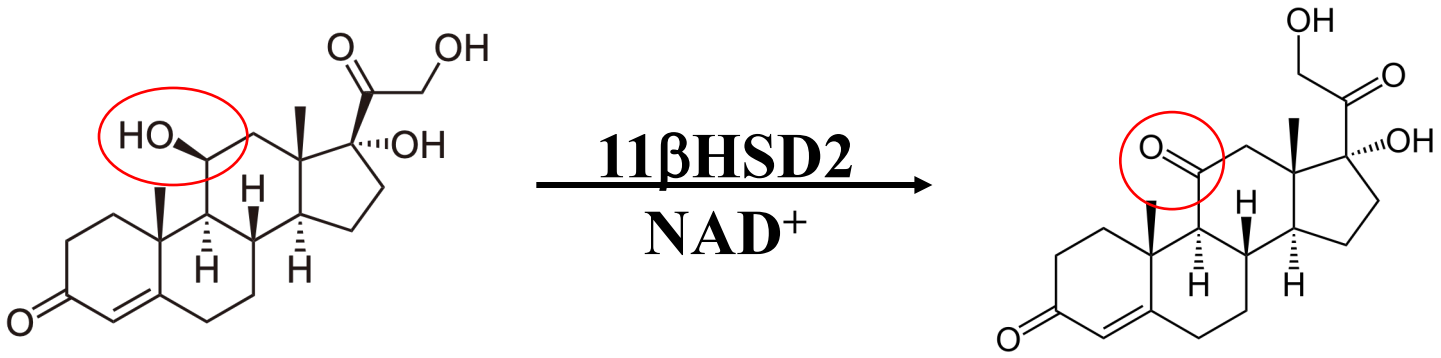
F. Pulmonary fibrosis

P value: 0.61

Tissue- and cell-specific effects of NHRs – example – mineralocorticoids

Tissue-Specific Steroid Modification in Aldosterone Signaling

- **Problem: Mineralocorticoid receptor (MR) binds equally well to aldosterone and cortisol, but cortisol levels are 1000-fold higher than aldosterone levels.**
- **Solution: To sense aldosterone levels, aldosterone responsive cells must first inactivate cortisol.**
- **11βHSD (type 2) catalyzes oxidation of cortisol into cortisone**
  - **Found in distal nephron (also placenta and colon)**

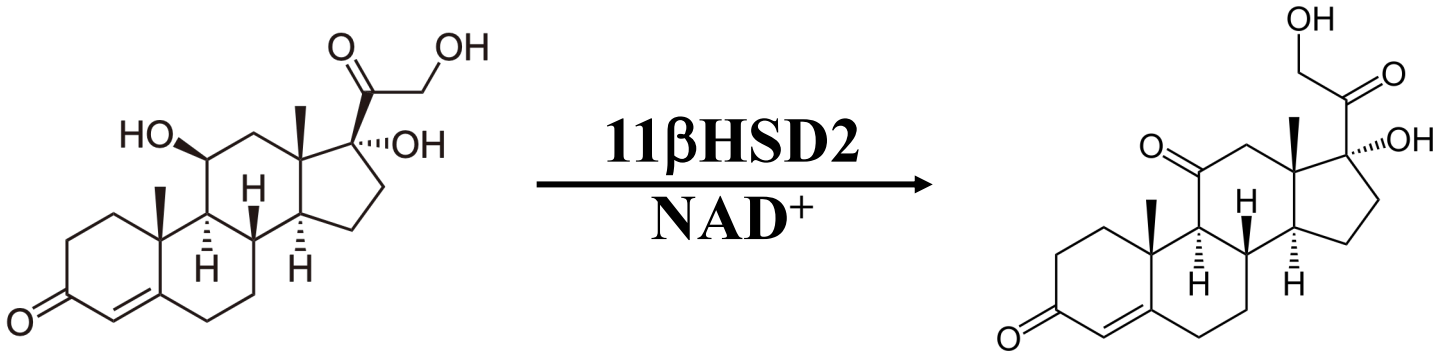


Cortisol (Active)

Cortisone (Inactive)

Disorders of Steroid Modification

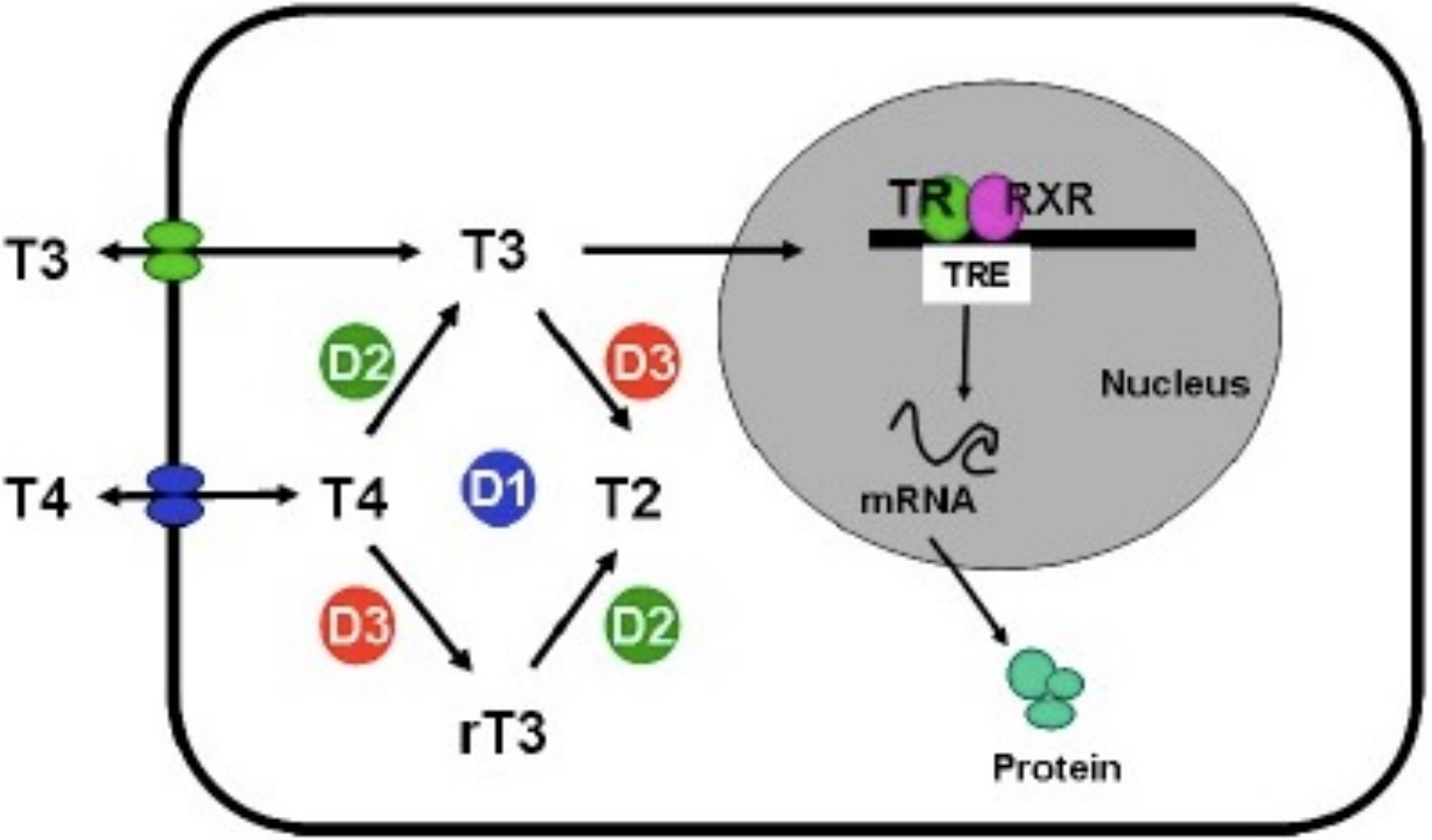
- Apparent mineralocorticoid excess (AME)
  - Caused by mutations in 11βHSD2
  - Symptoms: hypervolemic HTN, salt retention, hypokalemia
  - Labs: no mineralocorticoids, very low renin
  - Very serious condition - can lead to end-organ failure at a young age if untreated.
  - Treat with aldosterone receptor antagonists, low salt diet. Even with treatment, 10% die of CVA



Cortisol (Active)

Cortisone (Inactive)

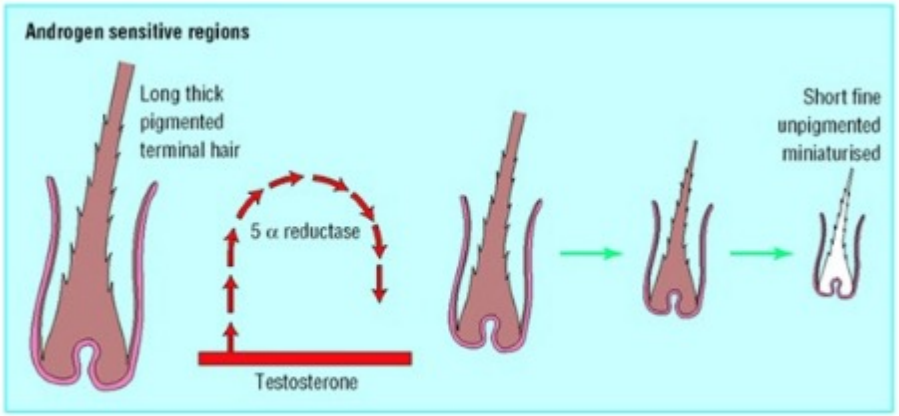
Tissue- and cell-specific effects of NHRs – example – deiodinases



https://www.ncbi.nlm.nih.gov/books/NBK285565/

Visser TJ. Cellular Uptake of Thyroid Hormones. [Updated 2016 Jun 15]. In: De Groot LJ, Chrousos G, Dungan K, et al., editors. Endotext [Internet]. South Dartmouth (MA): MDText.com, Inc.; 2000-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK285565/

Tissue- and cell-specific effects of NHRs – example – androgenic alopecia

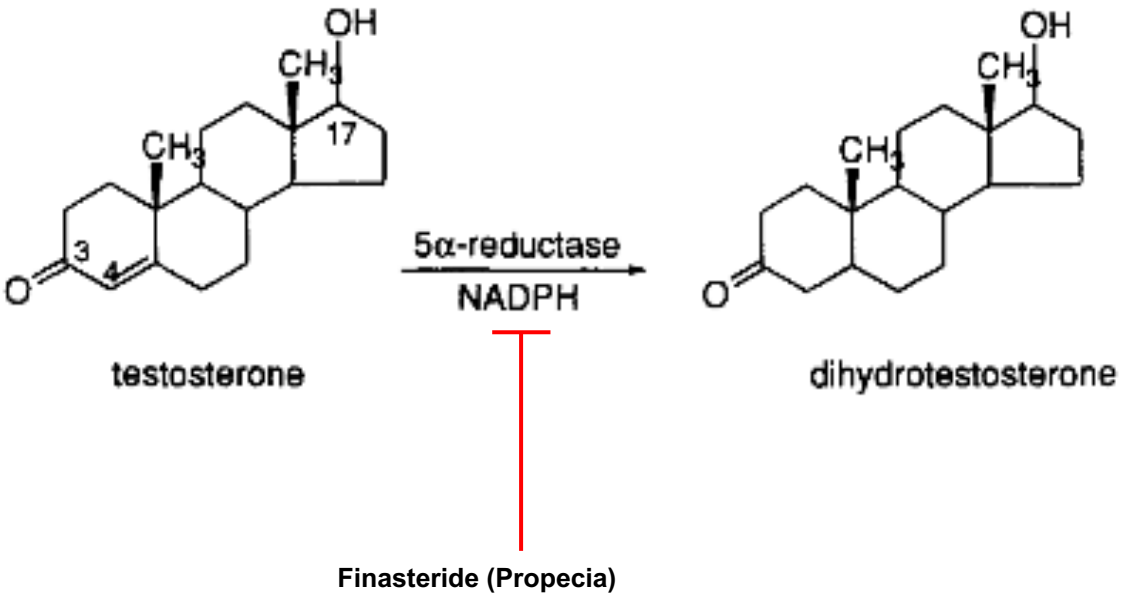


Stepwise miniaturisation of the hair follicle and shortening of the anagen growth phase, mediated by dihydrotestosterone



The growth phase of hair follicles is shortened by the transcriptional activity of **DHT**-bound androgen receptor (AR), but not by **T**-bound AR.

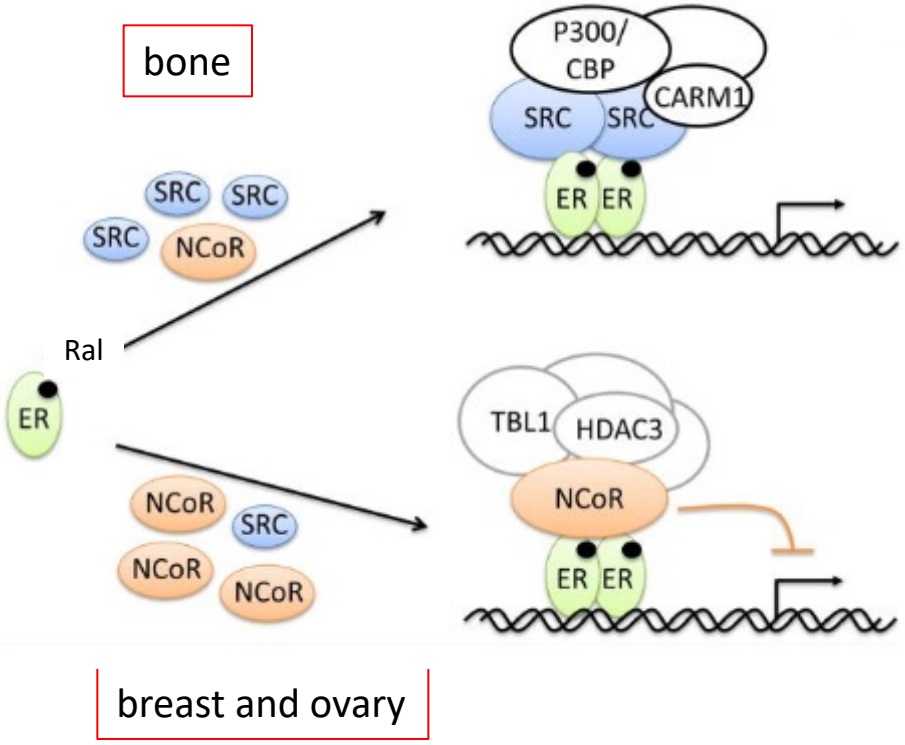
Tissue- and cell-specific effects of NHRs – example – androgenic alopecia



Inhibition of 5α-reductase prevents DHT-dependent miniaturization of the hair follicle, which stimulates hair growth.

Tissue- and cell-specific effects of NHRs – example – Estrogen receptor activity in breast cancer

- Binding to the ER by selective ER modulators (SERMs) – e.g., Raloxifene – results in tissue-specific ER activity.
- Changes the affinity of the ER for coactivators and corepressors – making ER activity dependent on the relative levels of these proteins in tissues.



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ScholarRx 2810.21

A research group engineered a mouse model to identify new treatment options for type 2 diabetes mellitus. A nuclear receptor called peroxisome proliferator-activated receptor (PPAR) is mutated to constitutively turn on gene expression irrespective of exogenous stimuli. It is then found that one class of medications is no longer effective in lowering serum glucose concentrations in the mouse. Which of the following is the most likely adverse effect associated with this class of medication?

A. Diarrhea

B. Disulfiram-like reaction

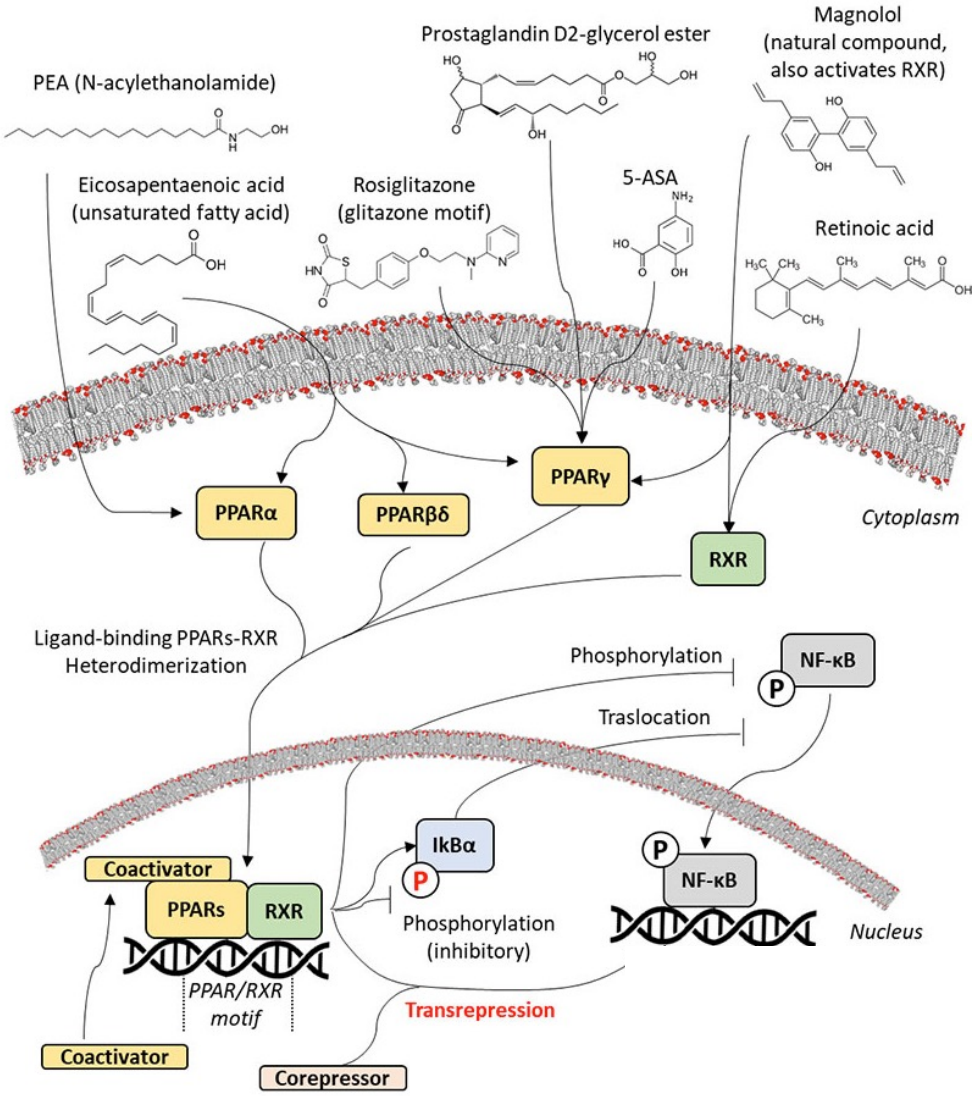
C. Hypoglycemia

D. Lactic acidosis

E. Water retention

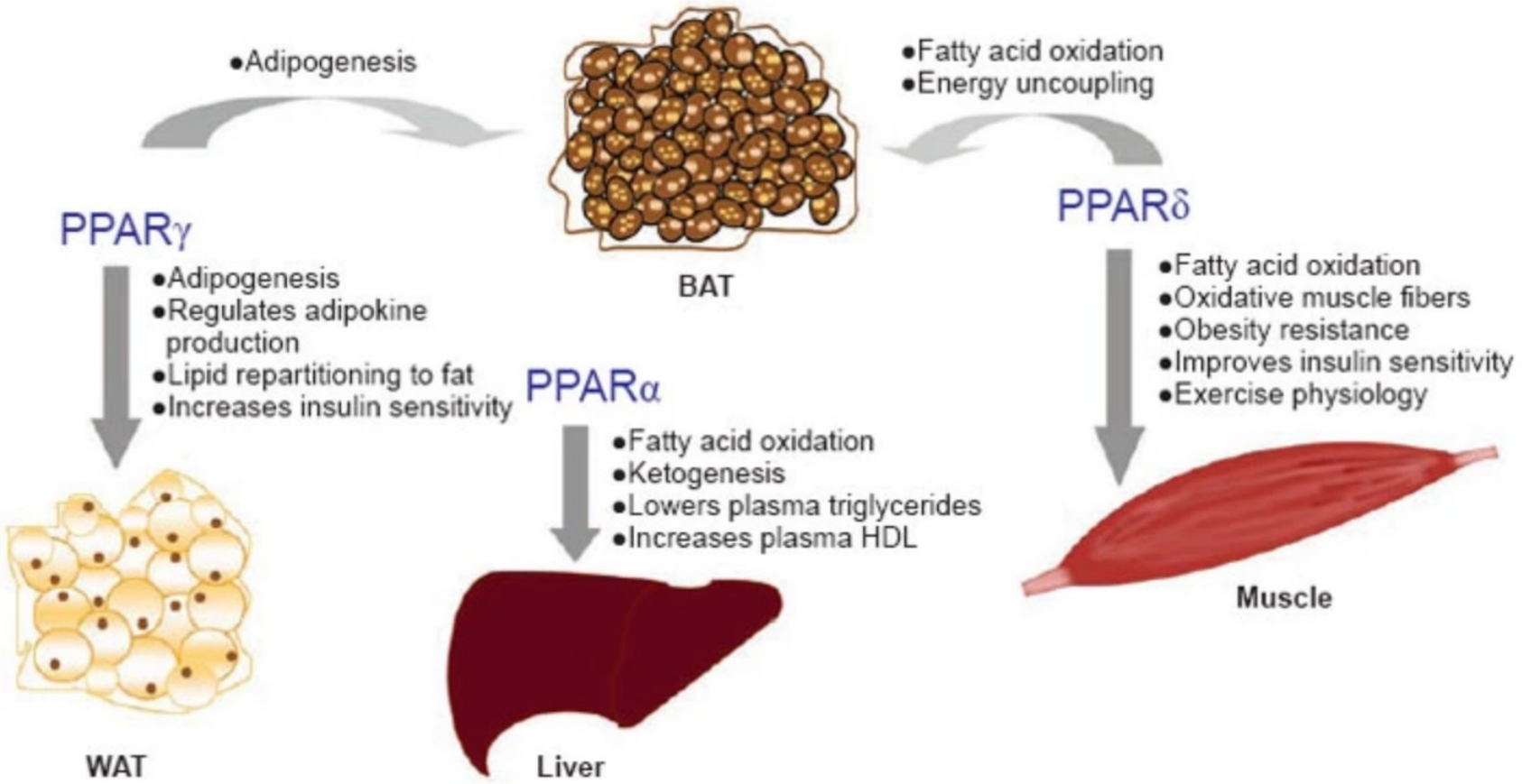
P value: 0.49

PPARs are cytosolic in the absence of ligands. They heterodimerize with RXR & translocate to the nucleus upon ligand binding.



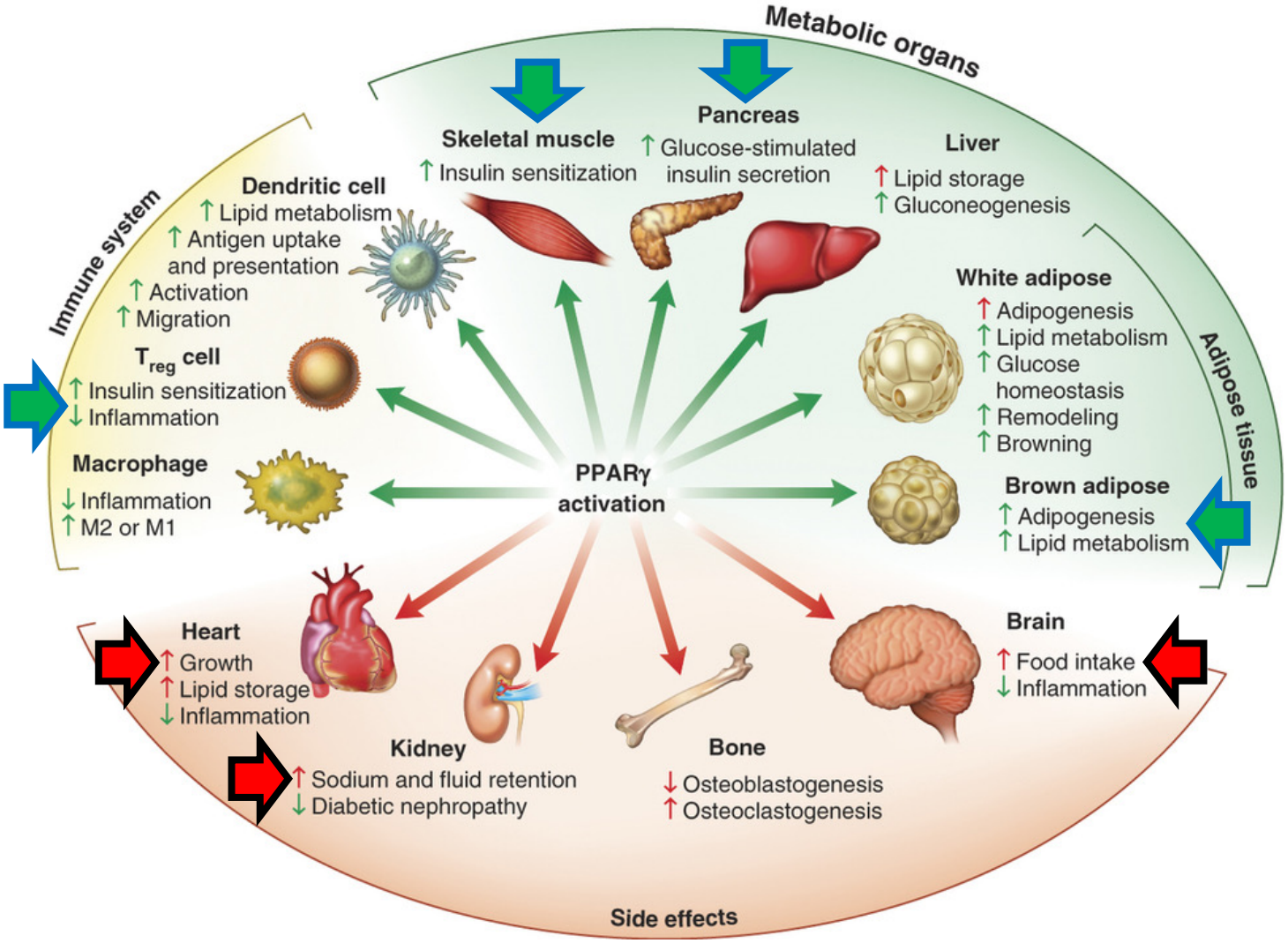
https://doi.org/10.3389/fphar.2020.00730

PPARs are important regulators of energy metabolism



Wang, YX. PPARs: diverse regulators in energy metabolism and metabolic diseases. *Cell Res* **20**, 124–137 (2010). https://doi.org/10.1038/cr.2010.13

PPARγ agonist benefits & liabilities



ScholarRx 2810.21

A research group engineered a mouse model to identify new treatment options for type 2 diabetes mellitus. A nuclear receptor called peroxisome proliferator-activated receptor (PPAR) is mutated to constitutively turn on gene expression irrespective of exogenous stimuli. It is then found that one class of medications is no longer effective in lowering serum glucose concentrations in the mouse. Which of the following is the most likely adverse effect associated with this class of medication?

A. Diarrhea

B. Disulfiram-like reaction

C. Hypoglycemia

D. Lactic acidosis

E. Water retention

P value: 0.49

Summary of Major Points

- Nuclear hormone receptors are intracellular, ligand-activated transcription factors
- NHR ligands may diffuse through the plasma membrane (e.g., steroids) or rely on specific transport proteins to access NHR targets (e.g., thyroid hormone)
- Type I NHRs cannot activate transcription in the absence of steroids because they are cytosolic; ligand binding stimulates dimerization and nuclear translocation
- HSPs stabilize poorly folded Type I receptors in the absence of their steroid ligands
- NHRs regulate transcriptional targets through activation, repression and trans-repression (tethering)
- Type II NHRs form DNA bound heterodimers with RXR and corepressors in the absence of ligands. Ligand binding stimulates corepressor dissociation and complex formation with coactivators, thereby activating transcription
- Tissue-dependent expression of ligand-modifying enzymes or transcriptional coactivators can influence effects of NHR ligands on target tissues
- PPARs display properties of both Type I & Type II NHRs